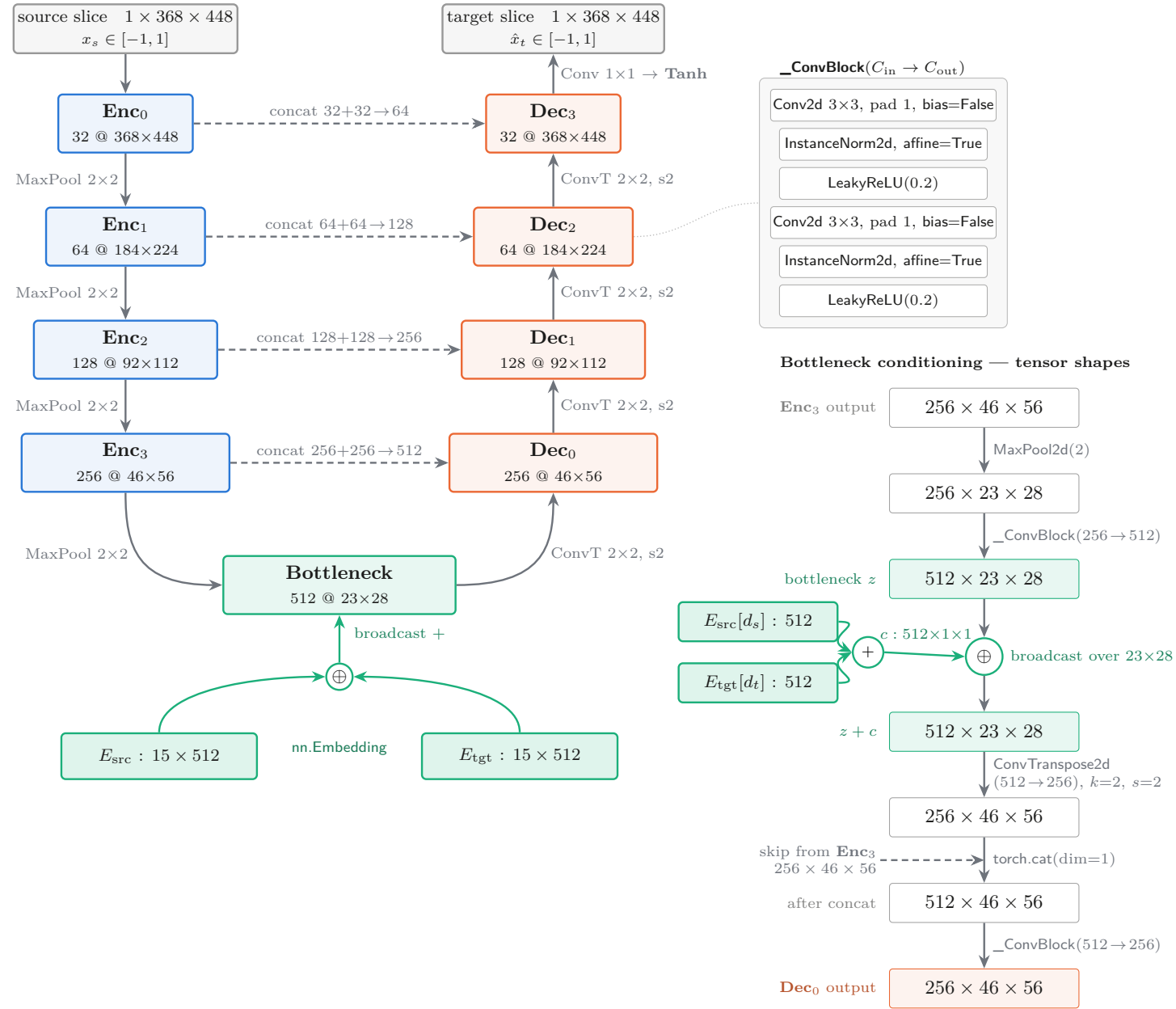


Conditional U-Net — Task 3 cross-field translation

task3_conditional_unet · levels=4, base_channels=32, max_channels=512 · 7.78 M parameters



$$d = 5m + f \in \{0, \dots, 14\}, \quad m \in \{\text{T1W, T2W, T2FLAIR}\}, \quad f \in \{0.1, 1.5, 3, 5, 7\} \text{ T}$$

$$c = E_{src}[d_s] + E_{tgt}[d_t] \in \mathbb{R}^{512}, \quad z \leftarrow z + c \mathbf{1}_{H' \times W'}$$

One shared generator covers all 60 ordered transfers (3 modalities × 5×4 field pairs); source and target always share modality.

encoder _ConvBlock

decoder _ConvBlock

bottleneck / conditioning

Block width encodes channel count on a log axis: $w = 22 \text{ mm} + 4 \text{ mm} \cdot \log_2(C/32)$. Heights are uniform and carry no meaning. Spatial sizes are printed, not drawn to scale. Hyperparameters on p. 2.

Task 3 conditional U-Net — hyperparameters

Run task3_unet_pro · checkpoint task3_unet_finetune_10.pt · config experiment-pipeline/configs/task3_unet_pro.toml

Architecture task3_conditional_unet

input_channels / output_channels	1 / 1
levels	4
base_channels	32
max_channels	512
num_domains	15
encoder channels per level	32, 64, 128, 256
bottleneck channels	512
normalisation	InstanceNorm2d, affine=True
activation	LeakyReLU(0.2)
output head	Conv2d 1×1 → Tanh
downsample / upsample	MaxPool2d(2) / ConvTranspose2d $k=2, s=2$
skip fusion	channel concat, then _ConvBlock
parameters	7 777 825 (7.78 M)

Data task3_paired

modalities	T1W, T2W, T2FLAIR
field strengths	0.1 T, 1.5 T, 3 T, 5 T, 7 T
domains	15 ($d = 5m + f$)
ordered transfers	60 ($3 \times 5 \times 4$; source/target share modality)
split	pro_train (include_retro=false)
crop_size	368×448
batch_size	4
num_workers	2
augmentation	random horizontal flip, $p = 0.5$
intensity range	stored $[0, 1]$, scaled to $[-1, 1]$
seed	0

Optimisation task3_unet_trainer

optimiser	Adam
learning_rate	1×10^{-4}
(β_1, β_2)	(0.9, 0.999)
epochs	10
amp	true (torch.amp autocast + GradScaler)
save_every	1 epoch
pretrain_steps	0 (no pretraining in this run)

Loss

task3_l1	weight 1.0 (F.l1_loss)
task3_lpips	weight 0.05, net="alex"
total	$\mathcal{L} = \ \hat{x}_t - x_t\ _1 + 0.05 \text{LPIPS}_{\text{alex}}(\hat{x}_t, x_t)$

Inference inference_task3_conditional_unet.py

axial slab	Z_CLIP_RANGE = [150, 180), 30 slices
grid	0.5 mm isotropic; output $364 \times 436 \times 30$
spatial handling	CenterCropOrPad to 368×448 , inverted after
intensity	$[0, 1] \rightarrow [-1, 1] \rightarrow \text{model} \rightarrow \text{clip } [-1, 1] \rightarrow [0, 1]$
size guard	F.interpolate (bilinear) before each concat
device	CUDA required (hard-fails otherwise)

Masked pretraining — variant task3_unet_retro_pro, not used by the run above

pretrain_steps	200 000
pretrain_learning_rate	1×10^{-4}
(β_1, β_2) , weight decay	(0.0, 0.99), 1×10^{-4}
mask_ratio / mask_patch_size	0.5 / 16
objective	L1 on masked pixels only, background (≤ -0.99) excluded
data	retro_train, unpaired, crop_size 512×512 , batch 2
subsequent fine-tune	50 epochs, lr 5×10^{-5} , $\beta = (0.5, 0.999)$, LPIPS weight 0.1